



LG Optimus 3D

Smartphones with 3D. Smartphones with 3D not only output movies and games in 3D using an autostereoscopic panel but they can also record and capture images in 3D. This kind of technology uses stereo image capture through two cameras placed a certain distance apart.

The different angles of the cameras with respect to the object being

photographed let the smartphone create a 3D view. As always, processing images in 3D doubles the load on the system. Hence an SoC that can handle a 20MP standard image might be able to handle only a 12MP stereo image.

LG's Optimus 3D utilizes TI's OMAP 4430. The OMAP 4430 is an ARM Cortex-A9 based dual-core SoC clocked at 1 GHz. It can handle 720p stereoscopic 3D video and capture 5MP stereo images.

An advancement over the OMAP 4430 is the OMAP 4460 featured in the Galaxy Nexus. The Galaxy Nexus is the current 'Google Phone.' The SoC that powers this device performs above the OMAP 4430 with a better full HD (1080p), 3D playback

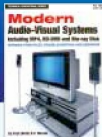
capability and also enough power to capture 12MP stereo images. This SoC (1.5GHz) is also clocked 500 MHz above the OMAP 4430 (1GHz). The Galaxy Nexus, however, does not feature stereo image sensors. Both the SoCs use PowerVRs SGX540 graphics core.

A smartphone that uses stereo image sensors to capture 3D images is the HTC Evo 3D. This device runs on Qualcomm's first dual-core SoC—the MSM8660. Clocked at 1.5 GHz and featuring an Adreno 220 GPU, this SoC is similar to the MSM8260. The only difference is that the MSM8260 is enabled with HSPA+ wireless communication, while the MSM8660 supports multi-mode HSPA+/CDMA2000 1xEV-DO Rev.B. The Adreno 220 has a theoretically slower fill rate than the SGX540 but the real-life performance difference is not known. ●

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